Phase 4 — Database Design

Source-grounded standalone document

Database Collections & Schemas

Entity	Fields	Validation / Constraints
User	_id, name, email, password, role, createdAt, updatedAt	ObjectId primary key; name/email/password required; email unique; password hashed; role defaults to user.
FAQ	_id, question, answer, category, createdBy, createdAt, updatedAt	ObjectId primary key; question/answer/category required; createdBy references User.
AI Generation (Logical)	topic, generatedQuestion, generatedAnswer, generatedCategory, generatedAt	Generated dynamically using Gemini before persistence.
Category (Logical)	categoryName, description, totalFAQs	Logical entity; totalFAQs is numeric.

Relationships

Relationship	Cardinality / Meaning
User → FAQ	One-to-Many: one user can create/manage multiple FAQ records.
Category → FAQ	One-to-Many: one category can contain multiple FAQ documents.
User → AI Generated FAQ	One-to-Many logical relationship.
FAQ → Search Results	Logical relationship for keyword/semantic search retrieval.